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PlayList : spring boot full course
channel : india free internship
youtube channel https://www.youtube.com/@IndiaFreeInternship
source Code: spring boot source Code
==
SESSION 1
----
1. SPRING FRAMEWORK
2. SPRING BOOT
3. SRPING DATA JPA
4. SPRING WEB MVC
5. SPRING REST
6. SPRING SECURITY
7. MICROSERVICE
8.SPRING BATCH
9.SPRING SCHEDULAR
======================================
SESSION 2
=========
Technolgies : webPage(servlets) +Database(JDBC/JPA) + Email (Java Mail) + Message(JMS)

Limitations:
- Start coding from line zero
- Development takes more time
- More chance of error
- Design pattern applied manually

Framework:
- Ready-made structure
- In-built technologies
- Design patterns already implemented

RAD => Rapid Application Development
- Fast coding
- Less Error

Spring Framework:
- used for Web Application or Distributed appliation
- It has own Apis
- It Follow Container System

Spring Container

-Find/Scan Classes[spring bean]
- Create Object
- provide DATA
-Link Object based on Relation
- Finally Destroy

Depedency Injection (DI)

Depedency Injection is a concept/theory which is implemented by spring IoC (Inversion of control) also called as spring container.

Theory	*programming*
oops	*core java*
ORM	*HIbernate with JPA*
DI	*Spring Ioc(Inversion of control)*

---

session 3
=========

Application
====================

3 Type of application

(i) Web Application
(ii) Distributed Application
(iii) Microservices Application

(i) Web Application : Collection of web pages run under webserver.

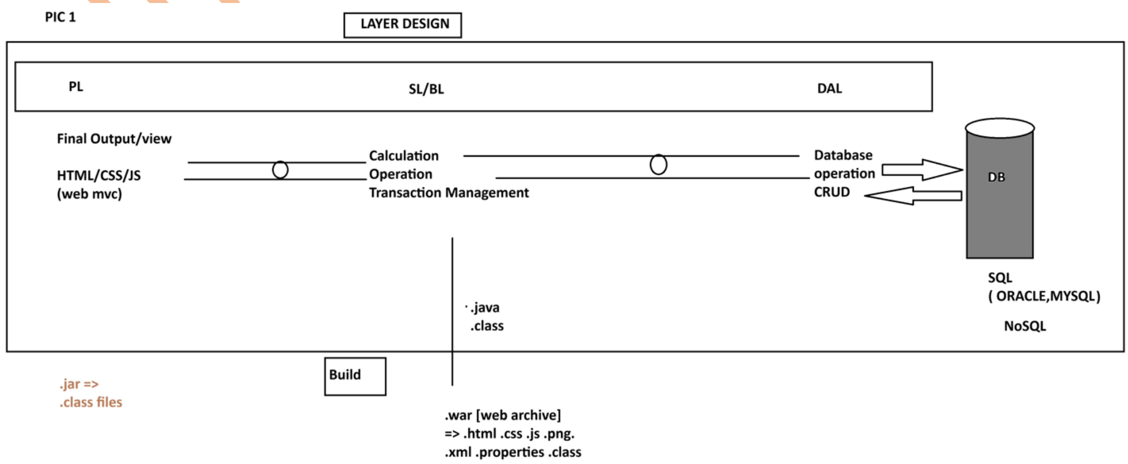
Application : - n services
Project :- n Module

eg : Gmail Application
User(Register and Login) , Inbox, Sent, Drfats, Setting, ... etc
Amazon Application
User(Register and Login) , search, Cart, Payment, ...etc

=> Implementation of service / module: - This is implemented using 3 Layers Design.

1. DAL : Data Access Layer (Database Operations)
2. SL : Service Layer (Calculation/Operations/txn mngmt)
3. PL : Presentation layer (view /Dynamic web pages/MVC)

PIC 1



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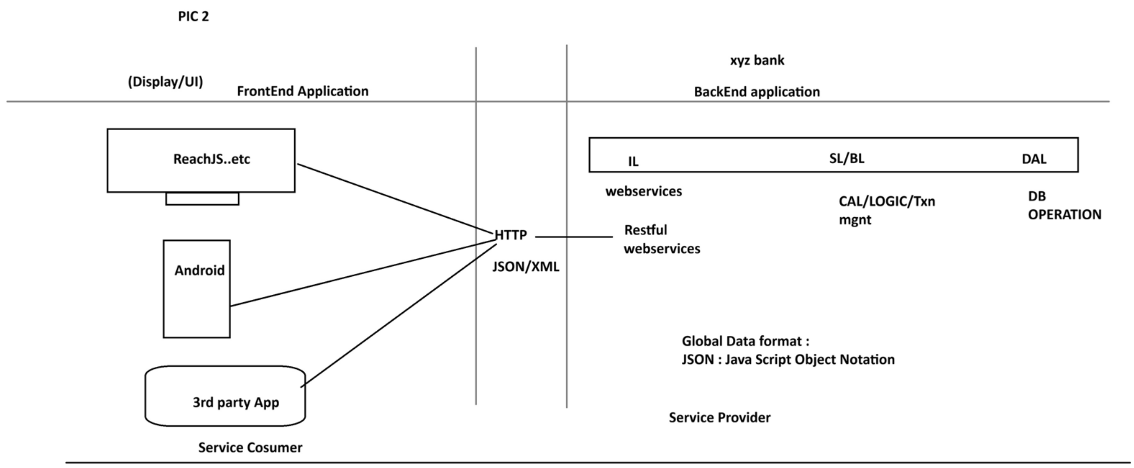
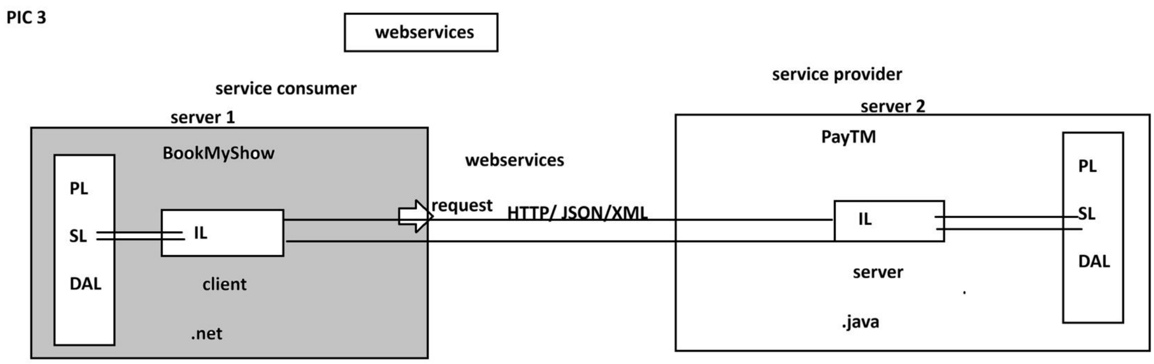
(ii) Distributed Application : An Application that runs at mutiple devices is called as distributed appliation.
xyz Bank application
This appliation can be accessed in different ways. Example.
eg:
1. Manually Process
2. NetBanking (Web App)
3. Mobile Banking
4. IVR
5. ATM / DC
6. 3rd party app (Gpay, PayPhone..etc)
.....

-But here business logics exist at backend application.
- Fontend applications are implemented using Angular, Android...etc
- App connected using HTTP Protocol using Global Data Format (JSON/XML)
PIC 2



WebServices:
===========
Intergration of multiple (at least two) applications which runs on different servers and implemented using different language (java, .net, php, python....etc)

PIC 3



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session 4

================
================
SPRING FRAMEWORK
================

CORE

----

1. *DI and IOC*
2. *Container Types*
3. *Annotation Configuration*
4. *@ComponentScan*
5. *@Component*
6. *@Value*
7. *Java Configuration*
8. *@Configuration*
9. *@Bean*
10. *@PropertySource*
11. *Environment*
12. *Lifecycle methods*
13. *Runners*
14. *@ConfigurationProperties*
15. *Properties file*
16. *YAML File*
17. *Profiles*
18. *Lombok API*

===========================
spring core chapter 1
===============

Core:
This module/chapter gives all rules and guidelines to work with spring container.
1. DI and IOC
============

Spring Container :- It takes care of object.
1. *Find/Scan our Classes(spring bean)*
2. *Create Object*
3. *Provide Data*
4. *Link Object*
5. *Destroy the Object*

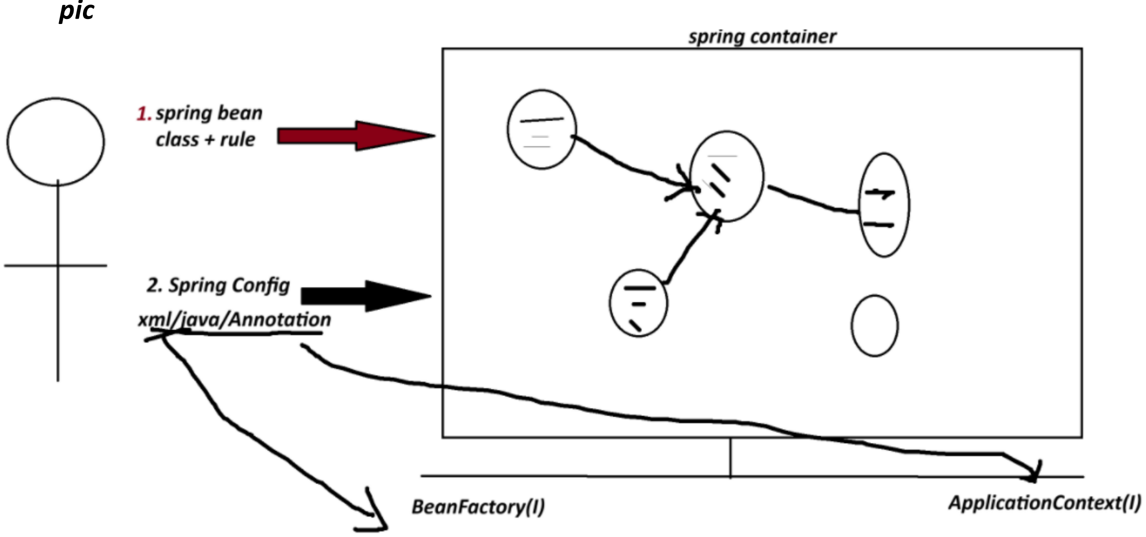
=>For this programmer has to provide two inputs-
1. *Spring Bean : class+rules given by spring container.*
2. *Spring Configuration : This is main input cotains details like object name, relation with objects, data....etc.*

xml/java/Annotation

*** Spring Container 2 Types:*
1. *BeanFactory (Old container) : support only XML coonfiguration file.*

2. ApplicationContext (New container) : support xml/java/Annotation configuration.

pic



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Dependeny Injection (DI)

Theory	programming
OOPs	java
ORM	HIbernate with JPA
DI	Spring IoC(Spring Container) -> Inversion of Control

=> Depedency :- An instance variable (Field) exist inside a class (spring bean)

1. Primitive Type Dependeny (PTD) :- If a variable is created using (8+1) datatype.
   (byte, short, int, long, float, double, boolean, char +string)
   (Their wrapper classess too)

2. Collection Type Depedency (CTD) :- If a variable created using(4) types:
   (java.util.package)
   (List,Map,Set and Properties)

3. Reference Type Depedency (RTD) :- If a variable is creatd using other class/interface

Injection(4)
======
=> It give/provide/assign existed provide data to Depedency(Inside object)

1. Setter Injection (SI) :- provide data to variable using its set method(setter).

2. Constructor Injection(CI): provide data creating object using parameterized constructor.

3. LookUpMethod Injection(LMI) :- Spring bean Scope
   [When parent bean is sigleton and child bean is prototype)
4. Interface Injection(II) : - Not Exist in Spring F/W.

```
class Employee{
  int id;
  String name;
  double sal;
}
```

id=~~0~~ 10
name=~~null~~ Raj
sal=~~0.0~~ 1200.0

```
Employee e=new Employee()
```

```
class Product{
  int id; //PTD
  String name; //PTD
  Boolean exist; //PTD

  List<String> models; //CTD
  Map<String,Integer> data; //CTD

  Vendor vob; //RTD
  Service sob; //RTD
}
```

```
class Vendor{}

interface Service{}
```

SESSION 5

===========

1. Spring Bean

2. STS Installation

Spring Bean

=========

-It is class that follow rules given by Spring Container.

Rule

-----

a. class must be public Type.

b. Recomended to keep in a package (basePackage Rule)

ex :

com.app.service

com.app.controller

c. Fields(Variables) are option, if required recomended to keep private.

private Integer empId;

private String empName;

d. If variable present in class,

=>define default constructor with setter/getter

=>(or/and) Parameterized Constructor.

ex:

```
public class Employee{
private Integer id;
private String name;
public Empoyee(){
}
 public void setId(Integer id){
this.id=id;
  }

public Integer getId(){
  return id;
  }
}
```

e. We can override Object(c) methods:

toString() equals() hashCode()

[because non-private, non-final,non-static]

f. Inheritance Rule:

Allowed ->Only Spring API is allowed (ie spring F/W classes and interface)

not allowed-> EJB, Servlet

g. Annotation Rule:

Only we can add Spring API Annotation and Inetegration API Annotations

(Hibernate with JPA, Restful webServices annotations);

ex:

@Component, @Service, @Repository.. etc.

STS Installation

=============

step 1:

go to official website

spring.io

link : - https://spring.io/tools#eclipse

step 2 download STS according your OS

step 3

Rigth click on ZIP file

click on "Extract Here"

step 4

Open extracted folder

Run : SpringToolsForEclipse

Final STEP STS installed successfully.

====================

session 6

=========

1. Spring Bean

Simple Java class that follow rules given by container.

2. Spring Configuration File(XML)

It gives object details (name, data,links with other objects ... etc).

3. Test class=> Just find, container created or not? object is creatd or not?

===

Spring core XML Tags

===

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

----

1. To create Object

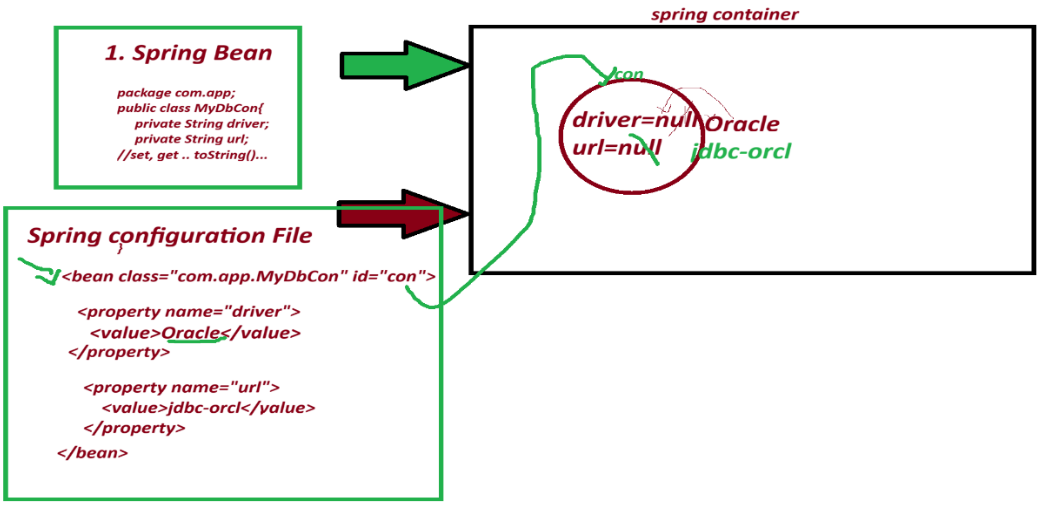
<bean id="objName" class="FullyQualifiedClassName"></bean>

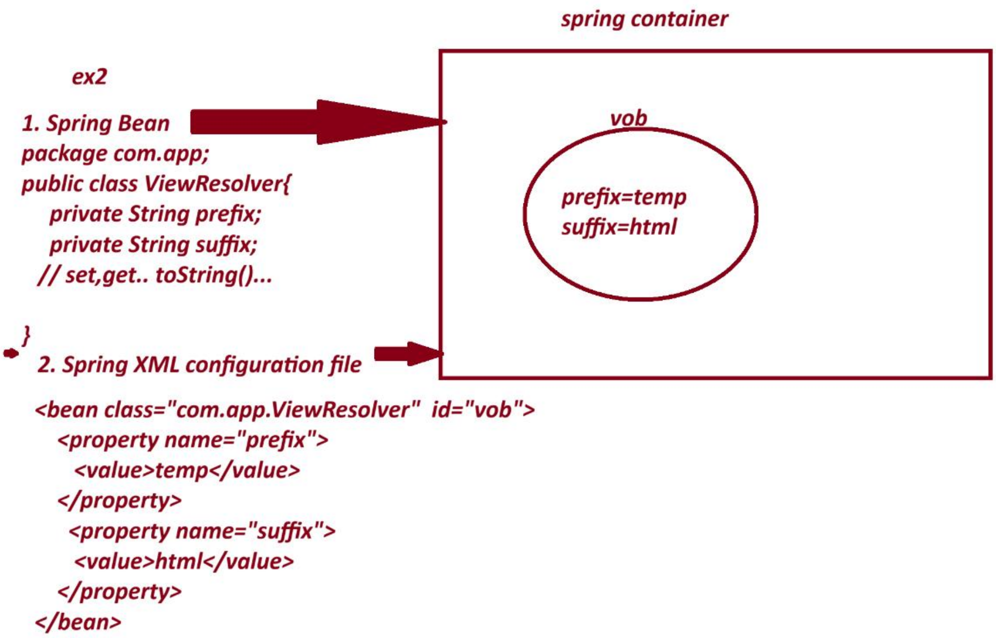
2. To call set method

<property name="variableName"></property>

3. To provide data to Primitive Variables

<value>data</value>





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1. Spring bean

```
package com.app;

public class PaymentService{

    private int token;

    private boolean active;

    private String provider;

   //set.. set toString..etc

}
```

2. Spring XML configuration file.

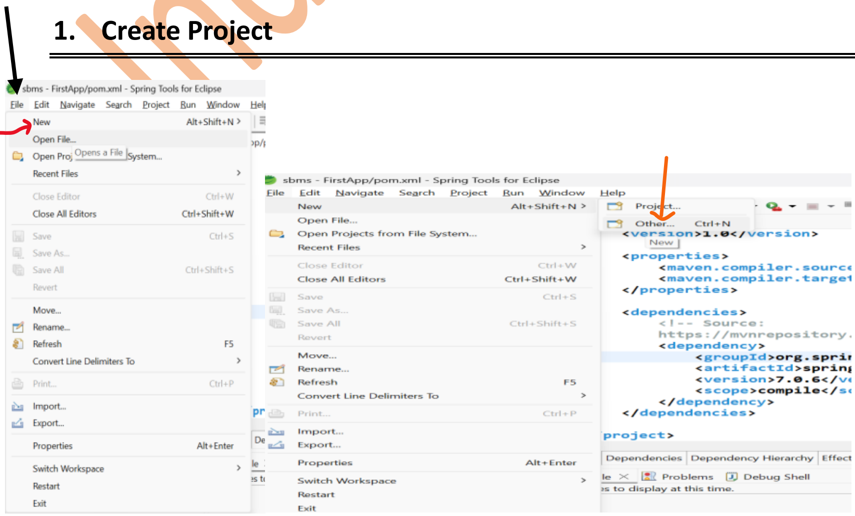
```
<bean class="com.app.PaymentService" id="psObj">
<property name="token">
<value>154678</value>
</property>
<property name="active">
<value>true</value>
</property>
<property name="provider">
<value>xyzBank</value>
</property>
</bean>
```

==

CREATE MAVEN PROJECT WITH BELOW STEP :-

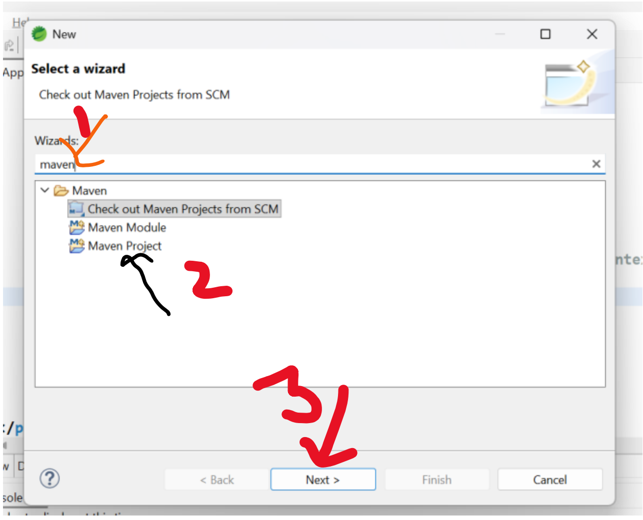
==

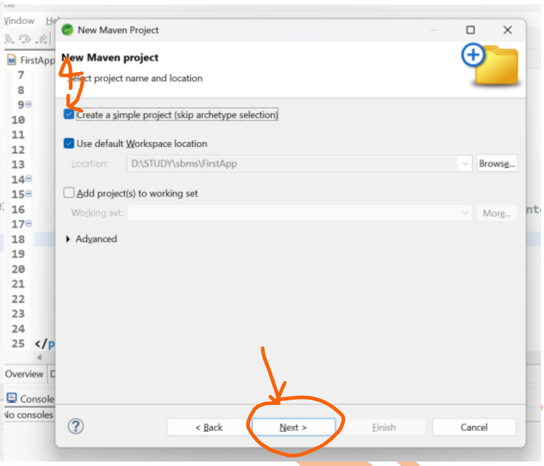
1. Create Project



File-> New-> Other-> Type maven->choose maven project-> next->

choose checkbox [v] create simple project -> next-> Enter details

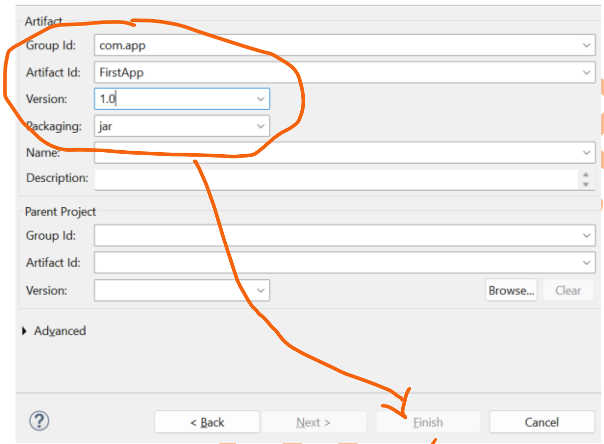




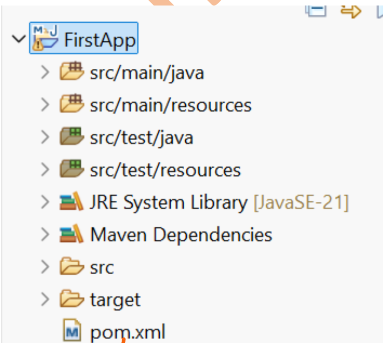
groupId : com.app

ArtficatId : FirstApp

version : 1.0



-> Finish



2. pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>FirstApp</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>

</properties>


<dependencies>

        <!-- Source:

          https://mvnrepository.com/artifact/org.springframework/spring-context -->

  <dependency>

    <groupId>org.springframework</groupId>

      <artifactId>spring-context</artifactId>

      <version>7.0.6</version>

      <scope>compile</scope>

  </dependency>

</dependencies>

</project>
```

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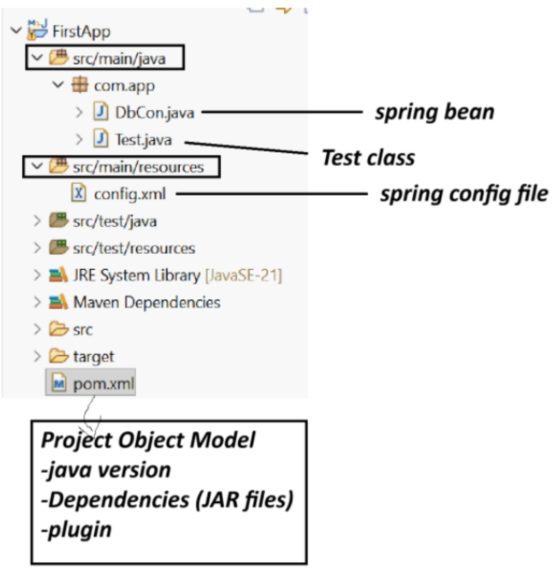
 ===========

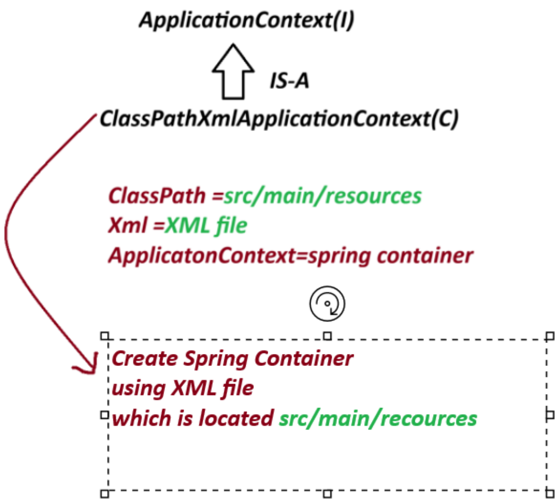
Spring Core First Application 3 Files.

1. Spring bean (.java)

2. Spring XML configuration (config.xml)

3. Test class(.java)





Note:-

1. Spring container are two Types.

(a) Legacy Container : BeanFactory(I) support only XML

(b) New Container : ApplicationContext(i) support all XML/java/Annotation config.

(c) ApplicationContext(I) : it is a interface.

we are using one of its Impl class:

ClassPathXmlApplicationContext(C).

==================================

Create Maven Project with below Step:

================================

File>new>other>type maven>choose maven project>next>choose checkbox> create Simple project>next>Enter details

GroupId - com.app

ArtficatId - SecondApp

version - 1.0

----

code

SecondApp

========

pom.xml

=============

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SecondApp</artifactId>
<version>1.0</version>

<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

DbCon.java

```java
package com.app;

public class DbCon {
  private String driver;
  private String url;
```

```
  //alt+shit+s
  public DbCon() {
super();
System.out.println("Constructor called");
  }
  public String getDriver() {
return driver;
  }
  public void setDriver(String driver) {
this.driver = driver;
System.out.println("Driver method called");
  }
  public String getUrl() {
return url;
  }
  public void setUrl(String url) {
this.url = url;
System.out.println("Url method called");
  }
  @Override
  public String toString() {
return "DbCon [driver=" + driver + ", url=" + url + "]";
  }

}
```

==

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
   xsi:schemaLocation="
       http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

   <bean class="com.app.DbCon" id="mobj">
       <property name="driver">
       <value>Oracle Driver</value>
       </property>

       <property name="url">
       <value>JDBC-ORCL</value>
       </property>
   </bean>

</beans>
```

======================

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
//ctrl+shit+O
public static void main(String[] args) {
ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
Object ob = ac.getBean("mobj");
System.out.println(ob);
}
}
```

----

output

Constructor called

Driver method called

Url method called

DbCon [driver=Oracle Driver, url=JDBC-ORCL]

---

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====================

Spring Container (DI - Depedency Injection)

=>Depedency - Variables exist in class

3 type

---

1. Primitive Type (8+1) : byte, short, int, long, float, double, boolean, char, String

2. Collection Type (4) : Set, List, Map and Properties

3. Reference Type (x) : using a class/interface variable

=> Injection : provide Data

- Setter Injection

- constructor Injection

-

LookUpMethod Injetion

- Field Injection (Annotations)

==

XML TAG

==

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

===

Collection Configuration

===

=> Spring container support working wiht collection configuration which are from java.util package

Collection Name	Collection Type	XML Tag Name
List	Interface	<list>
Set	Interface	<set>
Map	Interface	<map> and <entry>
Properties	Class	<props> and <prop>

List -> ArrayList

Set -> LinkedHasSet

Map -> LinkedHasMap

Properties --> NA

----

coding part collectionTypeS8

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>collectionTypeS8</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>
```

```
</properties>

<dependencies><!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean

MyDetails.java

```
package com.app;

import java.util.List;
import java.util.Map;
import java.util.Properties;
import java.util.Set;

public class MyDetails {
//ctrl+shift+O
private List<String> data;
private Set<String> models;
private Map<Integer,String> design;
private Properties myinfo;

public MyDetails() {
super();
```

```
}

public List<String> getData() {
return data;
}

public void setData(List<String> data) {
this.data = data;
}

public Set<String> getModels() {
return models;
}

public void setModels(Set<String> models) {
this.models = models;
}

public Map<Integer, String> getDesign() {
return design;
}

public void setDesign(Map<Integer, String> design) {
this.design = design;
}
public Properties getMyinfo() {
return myinfo;
}public void setMyinfo(Properties myinfo) {
this.myinfo = myinfo;
}
@Override
public String toString() {
```

```
return "MyDetails [data=" + data + ", models=" + models + ", design=" + design + ", myinfo=" + myinfo +
"]";

}

}

-------------------------------

xml config file

config.xml

-------------

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


<bean class="com.app.MyDetails" id="myinfo">

<property name="data">

<list>

<value>A</value>

<value>B</value>

<value>C</value>

<value>A</value>

</list>

</property>

<property name="models">

<set>

<value>M1</value>

<value>M2</value>

<value>M3</value>

<value>M1</value>

</set>

</property>
```

```
<property name="design">
<map>
<entry>
<key>
<value>101</value>
</key>
<value>AB</value>
</entry>
<entry>
<key>
<value>102</value>
</key>
<value>Suraj</value>
</entry>
<entry>
<key>
<value>103</value>
</key>

<null></null>
</entry>

</map>
</property>

<property name="myinfo">
<props>
<prop key="A1">B1</prop>
<prop key="A2">B2</prop>
<prop key="A3">B3</prop>
<prop key="A5"></prop>
</props>
```

```
</property>

</bean>

</beans>
```

Test File

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

public static void main(String[] args) {

ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");

//Object ob = ac.getBean("myinfo");

MyDetails ob = ac.getBean("myinfo",MyDetails.class);

System.out.println(ob);

System.out.println(ob.getData().getClass().getName());

System.out.println(ob.getModels().getClass().getName());

System.out.println(ob.getDesign().getClass().getName());

System.out.println(ob.getMyinfo().getClass().getName());

}

}
```

----

console output

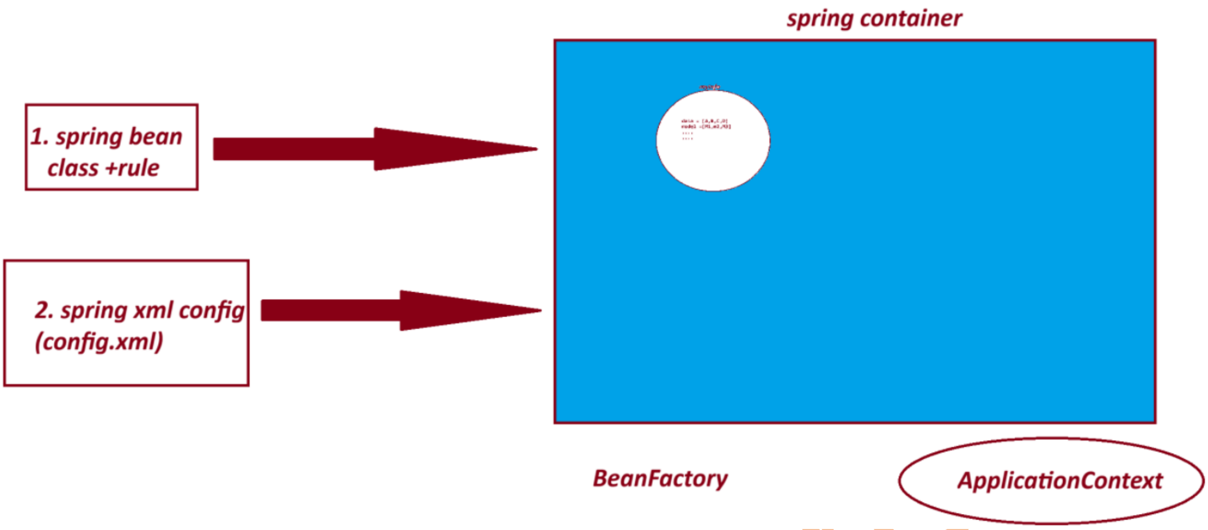
```
MyDetails [data=[A, B, C, A], models=[M1, M2, M3], design={101=AB, 102=Suraj, 103=null}, myinfo={A1=B1, A2=B2, A3=B3, A5=}]

java.util.ArrayList

java.util.LinkedHashSet
```

java.util.LinkedHashMap

java.util.Properties



===

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========

3. Reference Type Depedency - Setter Injection.

============================

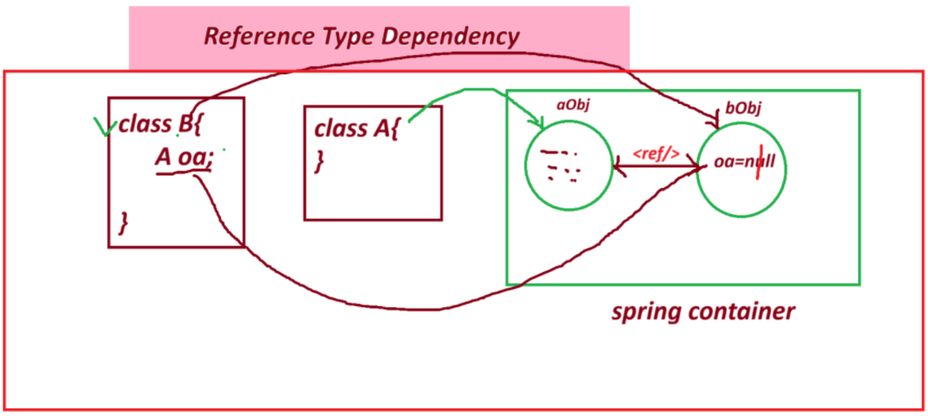
=> <ref/> : - tag is used to Link objects in case of HAS-A Relation.

Syntax:

<property name="variableName">

<ref bean="childObjName"/>

</property/>



======================

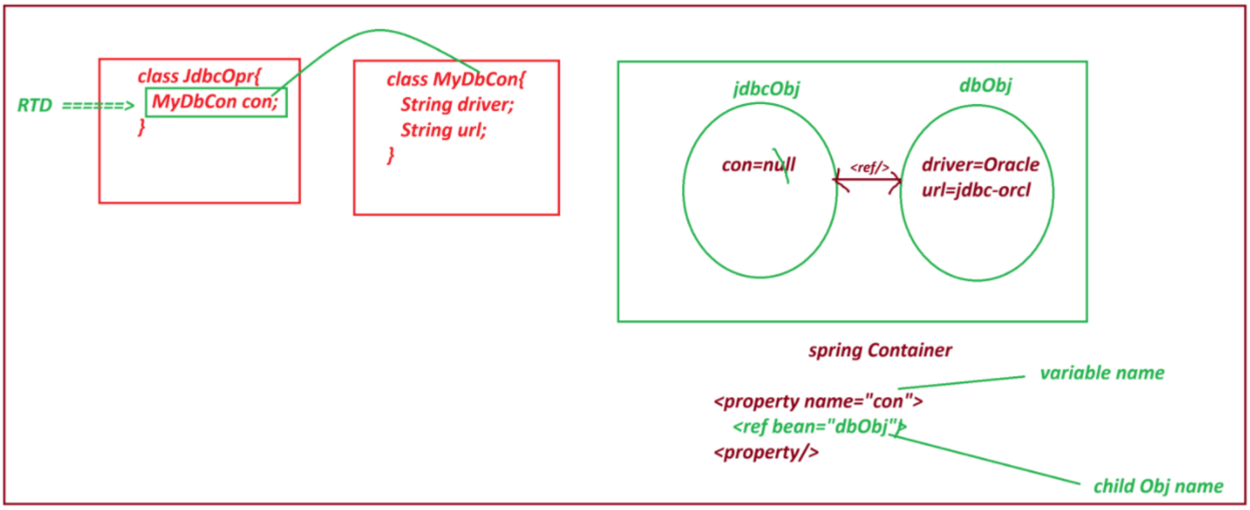
- If we do not use <ref/> then ref variable holds default value null.

=> spring container creates objet to both classes and link them by using <ref/> tag.

=> if child object nam (ref bean name) is not valid or not exist then spring container throws exception like

NoSuchBeanDefinitionException:

===



project : ReferenceTypeS9

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ReferenceTypeS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>
<dependencies>
```

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>

-----------------------------------------------------
JdbcOpr.java
----------------
package com.app;

public class JdbcOpr {
  private MyDbCon con; //RTD

  public JdbcOpr() {
super();
// TODO Auto-generated constructor stub
  }

  public MyDbCon getCon() {
return con;
  }

  public void setCon(MyDbCon con) {
this.con = con;
  }
```

```
  @Override
  public String toString() {
return "JdbcOpr [con=" + con + "]";
  }

}
```

===============

MyDbCon.java

```
package com.app;

public class MyDbCon {
  private String driver;
  private String url;

  public MyDbCon() {
super();
// TODO Auto-generated constructor stub
  }

  public String getDriver() {
return driver;
  }

  public void setDriver(String driver) {
this.driver = driver;
  }

  public String getUrl() {
return url;
  }
```

```
  public void setUrl(String url) {
this.url = url;
  }

  @Override
  public String toString() {
return "MyDbCon [driver=" + driver + ", url=" + url + "]";
  }

  //alt+shift+s

}
```

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean class="com.app.MyDbCon"  id="dbObj" >
      <property name="driver">
      <value>OracleDriver</value>
      </property>
       <property name="url">
       <value>JDBC-ORCL</value>
       </property>
  </bean>
   <bean class="com.app.JdbcOpr"  id="jdbcObj">
     <property name="con">
     <ref bean="dbObj"/>
```

```
    </property>
  </bean>



</beans>
```

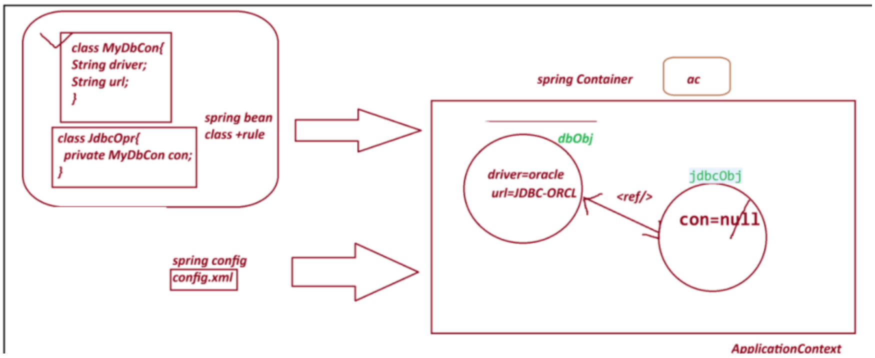
Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
public static void main(String[] args) {
ApplicationContextac=new ClassPathXmlApplicationContext("config.xml");
JdbcOpr jdbc = ac.getBean("jdbcObj",JdbcOpr.class);
System.out.println(jdbc);
}
}
```

output

JdbcOpr [con=MyDbCon [driver=OracleDriver, url=JDBC-ORCL]]

==



SI - Setter Injection :

Spring container creates object to given class using default constructor and provide data using setter method.

Employee emp=new Employee(); =======> <bean id="emp" class="Employee"/>

emp.setEmpId(10); =======> <property name="empId" value="10"/>

================================

*** Constructor Injection CI ***********

=================================

=> Sprint container creating object and providing data using Parameterized constructor.

Here , we use <constructor-arg> tag to indicate.

Employee emp=new Employee(10, "A"); =====>

<bean>

<constructor-arg>

<value>10</value>

</constructor-arg>

<constructor-arg>

<value>A</value>

</constructor-arg>

</bean>

Note:

A. If a class has 3 param constructor then we need to pass <constructor-arg> 3 times in xml config file.

B. Data must be passed in order using <constructor-arg> as per constructor difined in class.

==

code for CI (constructor Injection)

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ConstructorInjectionS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

Product.java

```
package com.app;

import java.util.List;

public class Product {

private String pcode;
```

```
private String model;
private List<String> data;
private Vendor vob;

public Product(String pcode, String model, List<String> data, Vendor vob) {
super();
this.pcode = pcode;
this.model = model;
this.data = data;
this.vob = vob;
}
@Override
public String toString() {
return "Product [pcode=" + pcode + ", model=" + model + ", data=" + data + ", vob=" + vob + "]";
}

}
----------------
Test.java
---------------------
package com.app;

public class Vendor {

private String vcode;
private String vaddr;

public Vendor(String vcode, String vaddr) {
super();
this.vcode = vcode;
this.vaddr = vaddr;
}
```

```
@Override
public String toString() {
return "Vendor [vcode=" + vcode + ", vaddr=" + vaddr + "]";
}


}
```

----

config.xml

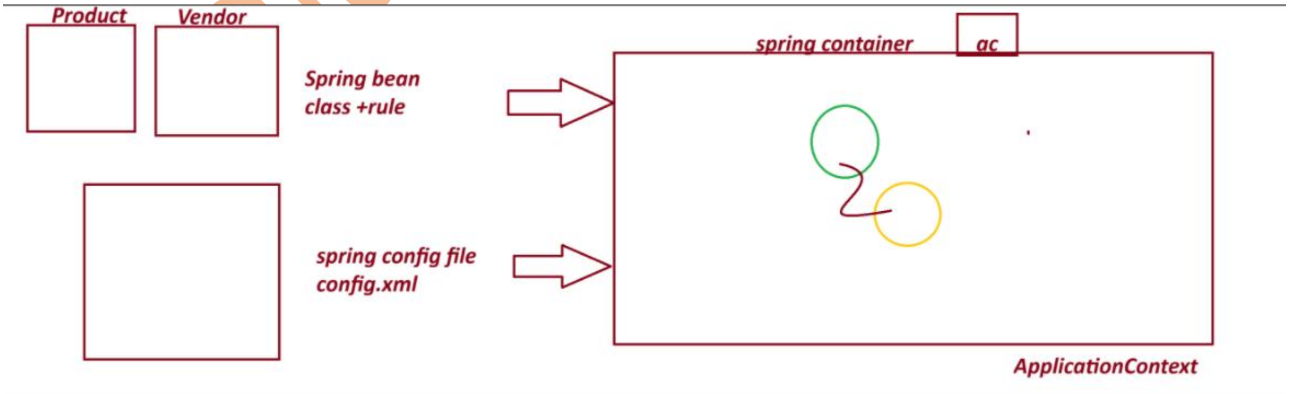
```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


 <bean class="com.app.Vendor" id="v1">
   <constructor-arg>
   <value>v1109</value>
   </constructor-arg>
   <constructor-arg>
   <value>IND</value>
   </constructor-arg>
 </bean>


 <bean class="com.app.Product" id="pobj">
  <constructor-arg>

<value>P01</value>
 </constructor-arg>



  <constructor-arg>
```

```
<value>HP-23444</value>
 </constructor-arg>

<constructor-arg>
<list>

<value>AA</value>

<value>BB</value>
</list>
</constructor-arg>

<constructor-arg>
<ref bean="v1"/>

</constructor-arg>
</bean>

</beans>
```

output

=========

Product [pcode=P01, model=HP-23444, data=[AA, BB], vob=Vendor [vcode=v1109, vaddr=IND]]



=====================================

session 10

===========

Spring Annotation Configuration

This is Easy to Configure

Faster in execution compare to xml.

======================

1. Stereotype Annotation:

These annotation informs Spring Container to create Object.

They are 5 Annotation given by Spring Framework.

@Component : Create object inside container.

@Repostitory : Creates object + Database operations

@Service : Creates object + Transaction/logic/calculation..etc

@Controller : Creates object + Http Protocol (Web App)

@RestController : Creates object + Http Protocol (webservices)

@ComponentScan

@ComponentScans

2. Data Annotation/ Basic Annotation

@Value Annotation is used to provide data (field inject) to variable.

It has 3 usages:

1. Hard coding direct value to a variable.

2. Reading Data from Properties/YAML Files

3. - SpEL (Spring Expression language)

@Autowire

@Qualifier

@Primary

==============

ExcelExportAnnotationConfigS10

code

========

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ExcelExportAnnotationConfigS10</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

--

Spring bean + annotation cofiguration

ExcelExport.java

```
package com.app;
import org.springframework.beans.factory.annotation.Value;
```

```
import org.springframework.stereotype.Component;
@Component("excel")
public class ExcelExport {
@Value("TEST")
   private String exportName;
@Value(".csv")
   private String fileExt;

@Override
public String toString() {
return "ExcelExport [exportName=" + exportName + ", fileExt=" + fileExt + "]";
}

}
```

MyAppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.ComponentScans;

@ComponentScan("com.app")
public class MyAppConfig {

}
```

Test.java

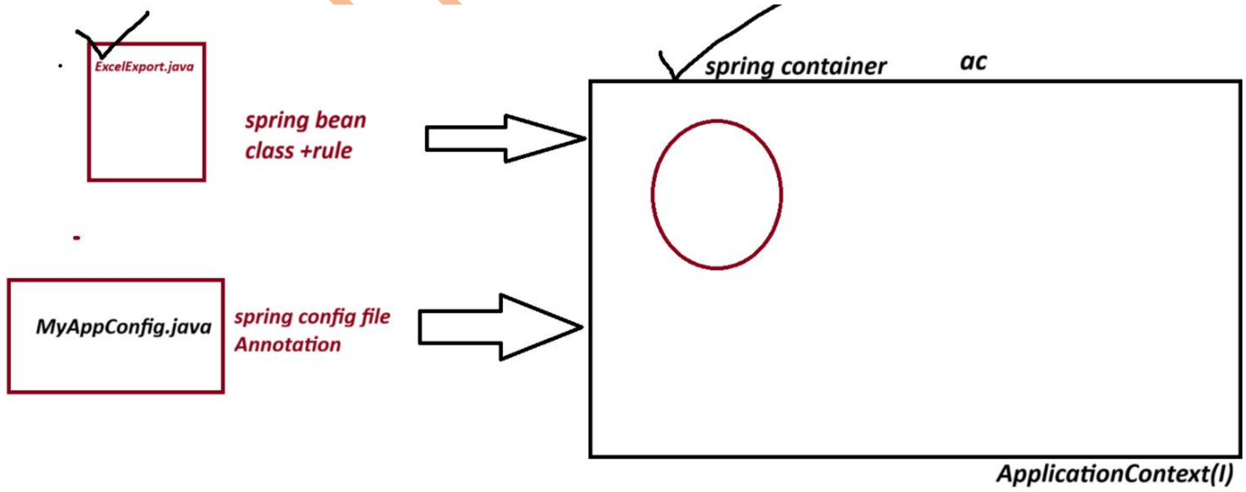
```
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
import org.springframework.context.ApplicationContext;


public class Test {
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(MyAppConfig.class);
        // Find classes from a package(basePackage)
//ac.scan("com.app");
        // object creation
       //ac.refresh();



        ExcelExport ee = ac.getBean("excel",ExcelExport.class);
        System.out.println(ee);
}
}
```

output

ExcelExport [exportName=TEST, fileExt=.csv]

Diagram



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Session 11